

CHAPTER 10

THE REALITY OF RACING

YOUR APPROACH TO A race should always be to do your best behind the wheel, but the level of excellence required and the risks you take to succeed change as you climb the racing ladder.

You approach the limit of your own abilities and the car's potential as closely as you can, but the closer you get, the finer the line is between a little mistake and a loss of control. Goal number one in your early races is to finish—it doesn't matter where—just finish and get car time in race conditions under your belt.

There will be races where you have to be right on the edge from start to finish in order to compete. There will be those where the level of intensity will be lower. It's easy to get carried away, so you should allow for at least one moment during a race weekend to remind yourself that this is most likely *not* the only thing that matters in your life.

QUALIFYING

Almost all types of racing establish the starting order of the race based on qualifying times. The fastest car sits in the pole and the cars are laid out two by two (or at some places, three by three), in descending order of their fastest lap times. Obviously, there are clear advantages to starting in the front. It's handy to not have to pass anyone—just hold your position from start to finish—but putting the car on the pole, or close to the front of the grid, takes on a different level of importance in different races.

Decisions, Decisions

A key factor in the importance of a good grid position is whether the racetrack is one with numerous good passing zones or one like a tight street circuit without any long straights, where it's going to be difficult to get by a lapped car, let alone a close competitor.

Another fact to remember is that in qualifying, in most cases, the grid position is based on one lap. Unlike the race, where a good average lap time can beat

Wheel-to-wheel competition creates additional mental and emotional demands on top of your existing set of skills and knowledge.

another driver who does an occasional faster lap but can't sustain it, doing well in qualifying only takes one good one.

The first consideration when approaching qualifying is to make an honest judgement about the importance of qualifying well in *this* race, at *this* racetrack. If it's one of your first few races, and you've got a lot to learn, it's not very important to do an all-out qualifying attempt. What you need at this point in your career is time behind the wheel of a racecar and you won't get that if you ball up the car in a banzai qualifying run. You may rightly decide to run qualifying somewhat conservatively, doing as well as you reasonably can, but not pulling out all the stops and risking your opportunity to get valuable race experience.

At the other extreme, if you're in a season-long chase for a championship and the pole earns points toward the title—and gets you the advantage of starting ahead of the rest—you and your team have to do everything you can to put in one blistering lap.

"In practice and qualifying you drive 100% the whole time, no—in qualifying you pick it up a little. In practice and qualifying at Long Beach this year I think I hit the wall six times, only doing damage once. Lots of times you brush the wall on the way out but that hasn't caught me out yet. You're just using all the road—if you've got a big power oversteer when you do it, obviously you're going to snap something. But at that level of racing and the seriousness of the competition, if you're not doing that stuff, you're not going to be on the front row. If you're leaving four inches from the edges during qualifying, you're not going to do it."

—Robbie Buhl

Car Set-Up

Car set-up for qualifying is necessarily different. For the race, the car has to be prepared to run for at least a half-hour, and in some cases up to 24 hours. In qualifying it has to go fast for a short period of time and this can mean that you change some systems on the car to give it a short-lived burst. To start, brand new tires are a must in qualifying because race tires have their ultimate grip in the first few laps of their life, then the grip stabilizes at a slightly lower level. There are all kinds of other tricks, from blocking off radiator and brake ducting, bypassing oil coolers, running thinner gear lube, and lowering fuel weight, all the way to installing a five-lap motor. The effort you put into tweaking the car for qualifying depends upon how important it is. It is often more important to spend practice time getting the race set-up right rather than playing with the ultimate one-lap set-up.

Your Driving Plan

From the driving standpoint, doing well in qualifying depends upon your ability to raise your driving to a higher level for a short burst. This takes intense concentration and a plan.

Frequently, using the draft of a teammate or a fast competitor can result in substantial lap-time gains.

"What I change for qualifying is that I'm more aggressive for a short-term period. I think that concentration is a big factor. I'll get tuned up to do three laps hard where I don't leave any cards on the table—I'll use the track up. I'll get really close to apexes.

"I consciously try to brake later in qualifying—really get the car in deep under braking. When you're talking six inches from the edges in measurements, that last six inches that you've left for a margin of error before nets you a qualifying time that picks up—whatever—two or three tenths, which you need in qualifying.

"Yet you couldn't race that way for a long time, leaving yourself no margin. If you take the margin for error out and you give it all you've got, I guarantee you're used up in three laps. You couldn't do any more because you've thrown all the cards on the table and cheated death. I don't even need to be told. I'll tell my guys before I even get to the start finish line, 'Get this lap, this is good,' and I'll get past start finish and get out of the throttle 'cause I'll know I can't do any more."

—Dorsey Schroeder

The draft has to be timed well. Two cars running in close company can both get a lap-time advantage since the two cars together will have less drag per car than each running alone, but sometimes this gain can be nullified for the following car if aerodynamic download on wings or ground effects is disturbed. The following car will get a straightaway drag advantage, but may suffer in the corners if the downforce is disturbed in the wake of the leading car. Using the draft, however, has the greatest advantage if the following car is not held up in corners by the car ahead, but can draft past the lead car on a long straight, getting to start/finish with a two- or three-car-length advantage relative to a normal lap. That three-car-length pop is worth .2 seconds at 120 m.p.h.

You have to weigh the draft advantage of following another car against the possibility of getting held up by the lead car. In some cases, it's better to find an open portion of the track to do a number of fast laps without the possibility of getting balked. This decision to tow or not to tow often depends upon the car: a car with relatively low horsepower relative to the amount of drag it creates will get greater benefit from drafting than a high horsepower car, slippery car, or one where aerodynamics are responsible for a good deal of their cornering grip.

REHEARSAL'S OVER

There are two fundamental problems drivers need to deal with when moving from practice sessions into an actual race. The first is the presence of other cars and their reluctance to allow you to pass, and the second is the mental pressure of the competition.

Traffic

In practice sessions, and even in qualifying, you always have the option to avoid traffic and find your own portion of the racetrack where you can operate freely, away from the influence of other cars. The fundamental problem in a race is that other cars may be where you want to be. Other cars will cause you to drive off-line in the corners, will affect your braking points, and will often determine when you can begin accelerating through and away from the corner. If you have set up reference points around the circuit to keep you on the optimum line and trigger your operation of the brake and throttle, a race situation is going to force you to adapt.

"Time trial" events, where only one car at a time is on the course, eliminate the traffic problem; but knowing you're on the clock cranks up the pressure just like a wheel-to-wheel race. It can be difficult to adapt to the conversion from the relatively low-pressure practice environment into the center stage of competition, where your performance is judged and rated by your finishing position.

Pressure

The tension is undeniable. You may have done hundreds of laps of a circuit, feeling perfectly comfortable and relaxed in the car, yet once you're in the middle of a field of other cars, you'd have to be on Thorazine not to feel pumped, especially in your first few races. The truth is that you'll never be able to totally control this heightened level of excitement, nor would you want to. Take the adrenaline out of racing and no one would do it. The trick is to not let it negatively affect your performance behind the wheel and your chances of beating as many of your competitors as you can.

Dealing with the pressure starts well before you get in the car and you can try a number of techniques to assure that the butterflies don't interfere with your hard-earned skills.

"My first professional race was a Barber Saab race at St. Louis and I remember I was extremely nervous. We went out on the pace lap and I forgot where I was on the grid. I was 10th or 12th—way back. When the pace lap started, my right leg started seizing up—I couldn't even push the throttle pedal, it was so bad. And the more I thought about it the worse it got—I mean, it was completely cramped up and frozen. I'm thinking, 'Oh no, we're going to come around to the green flag and I'm not going to be able to go 'cause I just can't push on the throttle!' I was just barely keeping up with the pace car and I'm thinking, 'Oh my God, I can't do this!'"

"Luckily, the guys at the front were so screwed up that the start was waved off and we didn't get the green flag. That was such a relief that suddenly my leg started to come back. By the time we got back around again my leg was back to normal again and I was able to go. That was kind of the last time I was ever nervous at the start."

—Jeremy Dale

Organize Your Time. The first step is to try to organize your time leading up to the race in order to avoid any last minute panic. Start by knowing the schedule of events and any specific rules and procedures which apply to the particular race event you've entered. For example, you may have a mandatory drivers' meeting preceding your race. Miss it and you may miss some important information from the officials—details like a changed start time for your race, or information about the track surface breaking up in a particular corner. You want to scrupulously avoid being late for any schedule deadline during the course of the event.

As the driver, it's your job to make sure that all your driving gear is available and ready to go. There's nothing more disconcerting than scrambling around at the last minute looking for a missing driving glove or having the visor of the helmet come off in your hand. Check over your stuff well in advance of start time and make sure nothing is missing or about to break. You wouldn't think that a broken shoe lace is any drama—until, that is, it happens three minutes before the start.



Fig. 10-1. Keeping all your driving gear in one spot and checking to see that it's all there, well before you have to get dressed, avoids the potential last-minute anxiety of searching for a lost glove.

Be certain of the time the car needs to be in the false grid area. Most road course events are run with multiple groups of races and as one race is underway, the officials will assemble the next group of cars in an area close to the pit lane called the false grid. They do this in order to get the next race underway without delay. In many events, you can lose your qualifying position and be moved to the back of the grid by arriving late at the false grid—an experience that isn't going to help your ability to stay calm.

Take Care of Yourself. You've got to remember to take care of your body too. Caffeine won't help the nervousness, so you might lay off the four cups of coffee you normally use to jump-start yourself in the morning. It also goes without saying that the dangers of alcohol and other mind-altering substances far outweigh any calming benefits they might provide.

If it's hot out, you'll need to keep plenty of liquids in your system as the race approaches. It's a guarantee that you'll sweat buckets anyway in the car, so water is crucial before you start. Consequently, you'll want to visit the bathroom shortly before you get in the car.

Confidence in the Car. You want to begin your race without any worries about the car's ability to perform. This is certainly more of a factor if you're transporting and prepping your own race car. Your race prep

checklist should include all the last-minute details that need to be attended to: most importantly, the simple things like tire pressure, fuel, oil, coolant, wheel torques, etc.

If a professional crew is prepping the car, it's worth asking the crew chief about their pre-race check procedure, just to give yourself the peace of mind that comes from knowing that the car is as ready to go as is humanly possible. Give the car a visual once-over yourself, looking at the suspension on each corner for loose jam nuts, disconnected sway bar links, missing safety wire—all of the little things that can go awry. Try to do this early, before the car is rolled to the false grid, so that in the one-in-a-thousand event that something is suspect, the crew can deal with it before it's crunch time.

Put Yourself in Neutral. Each driver has a slightly different approach to calming the butterflies. Jackie Stewart, one of the greatest Grand Prix drivers of all time, described the build-up of pre-race tension as akin to an inflating emotional balloon and his pre-race mental exercise was an effort to deflate the balloon—to try to become emotionally neutral so that his in-car decisions would be rational, not emotional. Remember that, after the first few corners, you'll be

"I don't know how to tell you to deal with race-day butterflies. I think everybody has them. I don't think your body or your mind can fathom going into a race, whether it's in a 230 m.p.h. Indy Car or a 110 m.p.h. Formula Ford, and not be nervous to some degree. Yes, there are people who are very confident in their ability, but still, racing is full of unknowns. It doesn't have to be your fault. There are 18 or 20 other racers out there that can take you along in their little deal.

"I think the best thing to do is not lock yourself away from everybody in the morning. If I leave myself alone for too long, then I get to think about it more, and the more you think about it the more nervous you become.

"I like to spend time with the guys and talk about what we're going to do and the strategy and all that's planned. Concentrate on the race, but from more of a technical standpoint than anything else. And then, 30 or 45 minutes before the race, I'll go spend some time by myself and run some scenarios in my mind—what I want to do and who's around me, what I can look for and what to gain, how to protect myself, and that kind of stuff. I think everybody has their own preferred method."

—Brian Till

back in an environment where you'll be too busy to consider the tension and it will vanish as you concentrate totally on the driving.

"I don't think anybody handles the butterflies the same....[B]efore I got into the higher levels with sponsors and everyone, I slept. I'd get nervous enough that I'd start to yawn—people thought I was bored—but the nervousness made me sleepy. I'd lay down and take a cat nap—doze away for 15 or 20 minutes."

—Danny Sullivan

STARTS

The most common type of start in U.S. road racing is the rolling start. Although each organizing group has slightly different procedures, typically you'll be given notice of the time until the start, three to five minutes before the officials signal you out behind the pace car. You stay in grid order around the pace lap as you perform your pace lap warm-up duties. As you complete the corner leading onto the start/finish straight, the pace car pulls off, leaving the pole sitter the responsibility of determining the pace of the start. The starter surveys the field and if the field looks orderly—that is, if everyone is in line and going approximately the same speed—the starter will wave the green flag and the race is underway.

If drivers are out of line or if the last rows are gaining on the front rows (or are past them) the starter will typically shake his head, "no," and continue holding the flag furled out of sight. A "No Start" can be hairy if some of the drivers don't get the message and go racing into Turn 1 as everyone else is slowing down. When it's an obvious "No Start," all drivers should get an arm in the air signaling to the cars behind that they're slowing down. When everything settles down, each car falls back into its original grid position and the pole sitter now controls the pace of the second pace lap.

Your Starting Plan

Be sure to check with the particular sanctioning body's rulebook and the supplementary regulations for the weekend so that you fully understand the start and re-start procedures being used, as well as the potential penalties for starting infractions like jumping the start. Conventions differ, and it's your responsibility to know the rules.

Be Flexible. When it comes to starting, preparation is everything. There is an important distinction between a starting plan that helps organize your thoughts and one that absolutely locks you into a single course of



Fig. 10-2. Gridded up and ready to go, the pace lap is about to begin.

"I think that at the start, you have to have a plan. Sure, when the flag drops and the race starts, everything is going to go out the window anyway, but at least you've put your mind in gear and made it start thinking about what you're going to do. You can't have an ironclad plan—you've got to adapt to what you get—but thinking ahead is worth the effort."

—Danny Sullivan

action. You can only go so far in planning specific moves to try to improve your position at the start, since it's likely that your vision of the start will be different from the reality. Deciding beforehand that you're definitely going to try to go outside to pick off the driver ahead of you might lead you to try it even when reality dictates that the opening isn't there. You need to be flexible enough to revise your starting tactics as the situation unfolds. Your plan should incorporate not only the routine aspects of all starts but should also take into account any factors unique to the particular race you're about to run.

Routine Factors. The routine aspects involve solving the standard problems which crop up when the green flag comes out. One is trying to race a cold car.

Taking a cold engine to its rev limit can be disastrous, so part of your plan should involve warming the engine on the false grid while you're waiting to go. Since many racecars have cooling systems that don't work very well when the car is sitting still, you have to be careful about not overdoing it—160 degrees of water temp will usually do. The pace lap will warm it up the rest of the way.

Once you're rolling, the next step is to try to get some temperature into the brakes, tires, transmission and differential. For the brakes, the most effective way is to get your left foot over onto the brake pedal and run the brakes against the motor, gently at first, then harder, to build the temperature in the pads and brake rotors. Since the tires are cold and the braking is biased toward the front when running the brakes against the motor, you have to be careful to avoid locking the front tires and flatspotting them.

The braking action can start to build heat in the tires and if you have a powerful enough engine, you can try to spin the rear tires to elevate their temperature. Most often you'll see drivers swerving right and left, scrubbing the tires with cornering force in an effort to get them warm. If you can do it, a long scrub is better than a lot of short ones, but traffic and the length of the racetrack will determine which method you can use. In any event, you'll never get them up to operating temperature this way, but it will improve

the traction from what it was when the tires were stone cold.

You can count on the fact that, depending on the type of car, it will take one to three laps at racing speed to get the tires to their optimum temperature; and you just have to deal with the limited grip until they get there. Most street tires don't need a significant warm-up to deliver their peak traction so there isn't much need to scrub them in on the pace lap, short of scrubbing the dirt off them.

You *do* have to be careful about all this braking and swerving on the pace lap, since many other drivers will be trying the same thing. If you get out of sync—for example, braking while the driver behind is accelerating—you can manage to crash out of the race before it starts.

"On the pace lap, I always left-foot-brake the car, and I'm surprised more guys don't do this. I don't keep it nailed down, but my left foot is always covering the brake pedal. I find that when you're scrubbing tires back and forth, the car is more stable with a little bit of brake pedal in it. It serves two functions—it stabilizes the car and it warms the brakes. That's about all I do on the pace lap, except maybe on cars that tend to load up a bit, I'll be sure to rev the motor to keep it cleaned out."

—Jeremy Dale

Be Observant. The pace lap gives you a chance to check out the track environment since it may be different than the last time you were on course. If it's a day of changing weather, be aware of cloud buildups, shaded areas of the course and the wind conditions. A 20-m.p.h. tailwind which wasn't there in qualifying can entirely change the nature of the braking zone at the end of the straight.

The pace lap will also be your first look in a while at the track conditions. Look closely, because you may encounter oil laid down in a particular corner or a huge collection of marbles (bits of rubber debris and dirt) off-line, built up from previous races.

Also get back into the routine of checking the critical gauges on your car, and expand your vision of the scene to include the peripherals, like corner stations and multiple rows of cars, not just the tail lights of the car directly in front.

Watch Corner Stations. We spoke earlier of trying to get a broader view—of avoiding the tunnel vision that occurs when you focus on the car or the road immediately ahead. This is especially critical when it comes to communication from the flag stations around the course. Seldom are they located directly in the line of

sight of the drivers, and it takes some effort at first to pick up flags in your peripheral vision. There is truly *nothing* more embarrassing—and no greater proof of tunnel vision—than missing the checkered flag which ends a practice session or race.

"You really have to watch the corner marshals. When they wave a flag, you have to appreciate that the flag is there for a reason. One of the first times I was in Mosport Park in Canada, a driver was killed in a Formula Ford because he came into a corner, racing, and ignored a yellow flag and ran under an ambulance that was parked around the corner for another accident. You can sometimes lose ground over it, but the flags are there for you and you have to pay attention."

—Danny Sullivan

Know Thine Enemy. Another important part of your pre-race plan is to know who you're racing against. Especially when you're new at it, you have to work at finding out who you can trust to run close with and which drivers are accidents waiting to happen. A standard part of your pre-race routine is to study the grid and identify the driver/car combinations that may present a problem. For example, if the driver starting next to you is fast but has spun out of the last six races, you might not worry so much if he gets by you early on, knowing that he's not likely to be there at the end of the race. If a certain driver has a history of dive-bomb passes, you'll know to take a second look in the mirror at corner entries when he's right behind you and to plan to set him up for the double pass coming out of the corner.

Studying the grid is especially important if you're racing in a field consisting of multiple classes. If a car in a faster class had mechanical problems in qualifying, problems which have been fixed for the race, that driver may be gridded behind you. After the start, you can expect that the car will be passing you based purely on car performance and that you don't have to waste time and energy defending your position against him. In fact, you may be able to tuck behind this car and make use of its draft to pull clear of competitors in your class.

Accordion Effect. One phenomenon you often encounter in starts is the accordion effect. You see it in street traffic every day. When the first car in a line accelerates, there is a bit of a time delay before the second car does, and on and on down the line, so that the field will tend to spread under acceleration.

The opposite happens under braking: the spacing between cars shrinks under deceleration. Depending

upon your position in the field, you need to consider how the accordion effect will influence your approach to the first few corners.

If you're back in the field, as you're likely to be in your early races, the most profound effect of this positioning on your driving will be the stack-up under braking for the first corner. If you're 8 or 10 cars back from the leader, you'll need to brake at least 8 to 10 cars earlier than you might have if you were running alone, just in order to keep from hitting the car immediately ahead.

"The biggest scare I ever had in a racecar was at the kink at Road America in a Formula 5000 car. I was on the pole and got beaten on the drag race at the start, and on the first lap I was a car-length behind the car ahead heading into the kink and he put the brakes on big time—a lot earlier than I was—and I nearly ran into the back of him. Could have been a 140-m.p.h. crash. AAAUUUGH—right up under his rear wing."

—Skip Barber

This is likely to feel really slow—and it is. The first lap of a race is often significantly slower than the later laps when the field spreads out—that's simply the way it happens. When you're new at it and focused on using distinct reference points for braking, the start will be the area where you need to go back to using your depth perception and judgment in traffic, just like you do on the street, but faster.

Elements of the Basic Plan

All of these aspects of your standard start plan aren't oriented toward taking any special advantage of your competitors. They're just your insurance that you have a car and driver combination which works reasonably well when the green comes out. To increase the odds of a good start, you can use this short list to prepare for every race start.

- 1) Assure that the car is completely ready, early.
- 2) Know your competition.
- 3) Get to the false grid on time.
- 4) Driver equipment accounted for and ready.
- 5) Warm motor.
- 6) Warm-up brakes and tires on pace lap.
- 7) Be observant.
- 8) Anticipate stack-up under braking for Turn 1.

Start Tactics

Now that you know the standard race problems are under control, the next step is to find a way to give yourself an advantage that can result in picking up positions early in the race.

When you're plotting a strategy for gaining positions at the start, you have to temper your plans with one irrefutable fact: *Few races are won in the first corner of the first lap but many are lost there.* If there are advantages to be gained without undue risk, by all means, take them. Just remember that the start represents a very small part of the race but one where there are a disproportionate number of car-to-car incidents.

The Green Flag Run. Much of the jockeying on the last part of the pace lap is an attempt to leave acceleration room ahead of you while at the same time trying to not let the driver ahead beat you to the accelerator



Fig. 10-3. In the early laps when cars are nose to tail, the brake point for heavy-braking corners like the Hairpin will be earlier than normal since you need to brake when the first car in line brakes, not where they brake.



Fig. 10-4. When the green flag comes out, try to take advantages where they exist, but remember that many more races are lost on the first corner than are won there.

when the green comes out. Meanwhile, everyone is trying to second-guess when the cars on the front row are going to floor the throttle and go. If it weren't so nerve-wracking, it would be comical. More often than not, you try to leave some room ahead and, as you back off the throttle to fade back, the field takes off and leaves you in the dust.

Your goal at the start is to pick off a car or two, if you can, and keep from being picked off yourself. In a class of relatively equal cars, if everyone begins accelerating at the same time, they should end up at the end of the straight in the same relative order in which they began. If you're anywhere behind the front row, you would like to time the start so that you leave enough room ahead to accelerate to a higher speed than the car in front of you. That way, you'll be gaining on the other driver at a closing speed of 50 m.p.h., still perfectly in line, just as the green flag comes out. That's your fantasy plan. In reality, a closing speed of two or three miles an hour would be sufficient. Then it's a matter of finding an empty piece of road to use to drive past the car ahead.

If you are able to get an advantage by getting a run on the car ahead, you now have to make a quick decision about which side to pass on. The mirrors are indispensable here, since someone may be gaining on you and an impulsive move with your eyes trained straight forward can cause big trouble. Sneak a quick peek before you pull out of line. Given the choice, you'd rather go inside—that is, on the apex side of the corner you're approaching. You can go outside, but you have to weigh the possibility of ending up being the only car outside the train that forms going toward the apex of the first turn. If the entire field ends up

nose to tail, you'll never get back into line and you'll lose many more positions than the one you've gained. Going toward the outside can work if there's a way to get back into the train, but it's risky.

Mind the Tach. The standard pace for the start of the race should be whatever speed results from being in second gear at an RPM in the middle of the engine's power band. In reality, it could be at any speed from a clutch-slipping crawl in first gear to as fast as you can go in fourth. Once the pace car leaves the field, it's up to the pole sitter, and he can choose whatever starting speed he thinks is to his best advantage. With cars that have relatively equal power, and especially with those that draft well, it may be an advantage for the leader to start the race very slowly in order to deny a significant draft advantage to the cars behind. Some drivers will choose to try to take advantage of the accordion effect and start full acceleration toward the flag as soon as the pace car clears the way, hoping that the field will string out before Turn 1. Although certain drivers fall into patterns, you can never be sure how the pole sitter is going to do it each time.

This makes it all that much harder to pick the right gear/rev-range combination that will give you the best acceleration run. In addition to sizing up the traffic and spacing yourself out correctly as you approach the start, you have to keep switching gears so that, if the flag comes out in the next instant, the engine is working in an RPM range where it's making some power. Once the race starts, you put your foot to the floor and go, but in all the jockeying and looking for holes, it's easy to forget that you have to shift it up to the next gear to keep from scattering the engine. You

soon realize that in normal laps, you very seldom look at the tach and follow the needle's progress toward the redline. Most often, you shift by the point of the track you're on or by the sound of the engine. At the start, however, you're at a unique portion of the track in an unusual gear; and you usually can't hear your own motor over the roar of the others. You have to keep sneaking looks at the tach as you approach the start and shift up when it hits the redline. Many an engine has been blown by this simple oversight.

Watch the Early Apex. The advantages of being on the inside of the first corner tend to make drivers rush to get there and, as we all know, if you're in early, the car will go wide at the exit of the corner and be slower than another car on the right line because of the tighter arc required late in the corner. Going in early could lead to making a pass on the way in, but losing it on the way out. It's even worse if you go in early and get alongside another competitor, but can't hold the car down to the inside because of your excess speed. As you drift up, the driver on the outside either gets pushed off the road at the exit or you hit—neither is a very attractive alternative.

Going into the corner early is, however, an acceptable strategy *if* you do it right. The value of beginning Turn 1 near the inside of the road rather than out at the turn-in point is both offensive and defensive. On the offensive end, you're better off being on the inside in tight traffic. As you saw earlier in our discussion of the line, when two equal cars corner side by side, the elapsed time of the inside car always beats the time of the outside car. In addition, if there is any leaning on one another (which we certainly don't condone, but it happens), the outside car usually gets the worst of it.

"In starts, there's a difference between being optimistic and being opportunistic. The biggest problem for the optimist is when Turn 1 is a wide corner, like Turn 1 at Sebring. You look up and there's all this virgin territory wide open around the apex, and you're tempted to take a straight short cut to the apex. You get there and realize that there are nine people headed toward that apex from the real turn-in point. You have to avoid that temptation."

"The opportunist is ready to take advantage of somebody else's mistake."

—Terry Earwood

On the defensive side, beginning the corner 7/8ths of a car width off the inside edge of the road effectively denies drivers behind you the opportunity to sneak inside on the way to the apex.

The key to making the early entry into Turn 1 work is to make sure to slow the car enough so that you can get back on line by the apex and have the car pointed in the same direction it would be on a normal lap.

Surviving Off-Line. Much of the overtaking going into the first corner is going to happen off the normal racing line. On some racetracks, the traction is comparable on-line and off, but in many situations, the grip will vary widely. It is very useful to devote some practice time to trying out different parts of the racetrack, especially the braking area on the inside of the road approaching Turn 1. It's possible that you may end up being forced to the outside of Turn 1 at the start, so it's a good idea to try running through the corner at progressively wider distances from the apex to see what



Fig. 10-5. At the start, remember that drivers who are able to stay on-line, single file, will have an advantage over those driving side by side through the corners.

it's like out there. Often you'll find that the grip is reasonable up to the point where the marbles make the track surface practically undrivable. Knowing acceptable off-line options will have a significant effect on your choices going into Turn 1.

Passing Positions. As you approach the braking zone for the corner, you have to make some quick decisions about your positioning. Can you get alongside one more car before the turn-in, or do you consolidate your gains and tuck into line? Will the driver coming up the inside yield when you go to the apex, or should you be the one to yield?

The conventions of passing hold sway as you approach the first corner. If you're inside and dead alongside or slightly ahead of another car at the turn-in point, they are obliged to yield. If you're halfway up, it's not good enough, and you should yield and pull in behind. If everyone in the field scrupulously followed these guidelines, the field would sort out quickly, but sometimes the driver on the outside will still try to beat you to the corner exit by staying outside. Sometimes a competitor gets a front wheel even with your rear and refuses to give up the two feet of road it would take for both of you to get back onto the racing line.

You want to keep in mind in during all this jockeying for position in the first lap that the drivers that are able to stay on-line are going to have a speed advantage over cars driving side by side through the corners. If you're stuck in a side by side duel, fighting for fifth and sixth place, there's a good chance that the first four cars may get away for good by staying on the racing line. Knowing when not to race—having the discipline to give up the chase of the moment for a chance of a higher finishing position later in the race—is the mark of a smart, seasoned racer.

THE BODY OF THE RACE

Once the drama of the start settles down, you can take stock of the situation. After a lap or two, you should have a good idea of any car or racetrack problems you'll need to contend with and plan how you'll adapt to the new situation. One or more corners may be more slippery than you anticipated and will take an adjustment in corner entry reference points or an adaptation of line. In cars with adjustable brake bias or sway bars, you may have to make a change in trim to improve the car.

You'll also get a good idea of how the field is shaking out so you can narrow down the number of competitors you'll have to deal with. The qualifying system, if it works correctly, should lump together drivers and cars that have been turning laps in the same range of lap times. Whether you're gridded 3rd or 23rd, you can expect to compete with the drivers who qualified a little faster and a little slower than you: this group can consist of 2 cars or 8 cars.

For drivers of all skill levels and experience, the gratifying fact is that racing with three cars for the lead of a race is just like racing with three cars for 10th. Your approach will vary in races with different types of cars and different durations, but many considerations apply to all races, whether they be 24-hour endurance events or 20-minute sprints.

"If you ask most racecar drivers about what was their best race, often it's not the one that was the best result. Some of the most heated battles I've had were for 7th or 8th place. I remember the South African Grand Prix in 1983 where I had to start the Tyrell from pit lane because we had some last-minute car problems. In the end, I finished 7th, and for me, it was really satisfying."

—Danny Sullivan

Seeing Independently

As you saw at the start, your well-drilled reference points around the racetrack have sometimes got to be abandoned and replaced by your best guess and estimate of the path and speed that will work. Once traffic thins out, it gets easier to pick up the routine, but even if there is only one car in front of you, you can't see ahead as well as you did in practice when, for the most part, you were running alone.

"You've got to be careful not to just copy what the car ahead is doing, especially if you're coming up on a slower car. There are two possibilities. He's either watching you too closely or not watching you at all. If he's not watching you, he'll drive you off the racetrack if you give him the chance. If he's watching you carefully, more often than not, he'll miss the turn-in and end up wide of the apex. If you're following him too closely, it's easy to do the same thing. I really try to look through the car and get an overall feel for where I am, and get by on the path I choose."

—Jeremy Dale

During a race you have to work hard at looking past the car ahead to pick up your braking and turn-in points. It's very easy to follow the driver ahead and get caught up in doing just what he or she does. You have to remember that your goal is to do a better job than the driver ahead and you certainly don't want to end up repeating anyone else's mistakes.



Fig. 10-6. Finding your reference points is more difficult when nose-to-tail with other cars. You have to look past the car ahead, rather than just mimic what they do.

To Pass or Not to Pass

As a general rule, when the opportunity to pass the car ahead presents itself, take it. There are a few things to keep in mind, however. If the pass puts the two of you side by side through a corner or two, you'll both have to go slower than you could have by staying on line through the turns. Driving side by side, lap after lap, will let drivers ahead get away, and let drivers behind catch up.

If possible, you would like to complete the pass and get back in line for the corner. Ideally, you would then pull away from the driver you just passed and set off after your next victim. In many classes of car, however, the draft effect is so significant that the car you just passed can stay with you, using your draft, even if you are slightly faster through most of the corners. If the other driver stays with you but can't seem to muster a draft-by, it's unlikely that there's much chance of



Fig. 10-7. Mistakes create opportunities. Here, car No. 3 has gone hopelessly wide and will be gobbled up on the way to the next corner.

being overtaken unless the driver behind tries some kind of desperate move. Keep checking the mirrors as you approach the most likely out-braking sections of the course so that if the other driver tries the dive-bomb, you won't get taken out. In a lap or two it will become apparent whether the other driver is much of a threat or just a drafting aid that will help you claw closer to the front.

The car behind is hoping for you to make a mistake—overreaching on your brake point, getting wide of the apex, or early-apexing a corner—and will take advantage of any bobble. Your job is to concentrate on driving as well as you can and to try to improve your lap time using the old Line, Exit Speed, Braking progression.

"Anytime, whether it's the third lap or the third to last lap, if there's a car in front of me, I want to catch that guy and get by him as quickly as possible. The fact that it's close to the end of the race doesn't add any sense of urgency to me....[although] I suppose it depends on the situation a little bit. If it's the last lap of the Indy 500 and you've got a shot, you're going to take it, almost regardless of what the risk is."

—Bryan Herta

Meeting Your Match

There are occasions when you'll be so evenly matched with another competitor that you'll be able to draft by each other on the long straight, every lap. You'll lead a lap, then get passed in the braking zone. He or she will then do the same. If this sort of scenario is developing, you can make the choice to keep swapping positions, to work together to get better lap times for both of you (either to catch the drivers ahead or leave the rest behind), or to settle back and follow, plotting a strategy of how to wind up in the lead on the most important lap—the last one.

Working together requires the active participation of both drivers. Once you've felt each other out and it becomes obvious that neither one of you has the capability of pulling away from the other, you can try to communicate with the other driver your willingness to work as a team. As the other driver drafts by, you can signal by first trying to make eye contact with the other driver, then pointing straight ahead, essentially saying, "Let's go." Using your full hand, karate-chop style, also conveys the meaning, "Let's stay in line." You're trying to arrange it so that both of you can take advantage, on successive laps, of the extra straight-away speed you get from the draft without losing the lap time advantage. You don't want to arrive at the entry of the turn at the end of the straight side by side,

forcing both of you off-line in the first part of the corner. The driver who's getting passed may need to breathe off the throttle toward the end of the straight to allow the other to get completely by so that both of you can run through the turn on-line. At some point, four or five laps from the end of the race, the cooperation will end as each of you try to maneuver your way into finishing ahead of the other on the final lap, but the pair of you can end up closer to the front by calling a temporary truce.

If you can't get the cooperation going or if you're in a race of your own by being far ahead of any others and pulling away (or steadily losing ground to cars at the head of the field), you may choose not to pass the other driver, even though you could. Following can do two things: it tends to put more pressure on the driver ahead, making him or her more prone to mistakes, and it allows you to survey the other driver's relative weaknesses and strengths.

Watch and Learn

You're hoping to find a corner or two where you are clearly faster, not enough to pull alongside or pass, but a speed at which you would pull out a few car lengths if you were ahead. This knowledge can be useful later in the race when you come upon other traffic. If the two of you are approaching a lapped car, you could try to maneuver yourself into a position where you're leading your adversary as you approach traffic. Try to adjust your pace so that you can pass the lapped traffic following your fast corner, on the chance that your adversary might not get up enough steam to get by the lapped car before the next corner. You'll pull out some distance in your good corner and more distance if your competitor is balked by the lapped car in his better corner.

Another advantage of following is that you can go to school, learning how your foe manages to go quickly in corners where your speed is lacking. If you adapt your technique to mirror his, and over the course of a few laps manage to match his pace in the corners where you were lacking, you might combine your new-found speed with your advantage in other corners to be able to pass and pull away. Once you feel that you know some new tricks, go ahead and take the lead and you'll have one lap, two at most, to try to put the lessons to use and pull away enough to break the draft (and sometimes your rival's spirits). If you make mistakes on your killer lap, the other driver will be able to keep up and then go to school on you; so, to make this work, you've got to do the best lap you can manage, or you're back to square one.

THE FINAL STAGES

When it comes down to the final few laps, the game becomes one of maneuvering yourself into a position to arrive at the finish line before your competitor on the last lap.

The order on the previous lap doesn't matter, and in many cases, it's better to be behind your competitor on the next to last lap than ahead. If the draft is a big factor, you may be the victim if you're leading on the run to the finish line. There's really nothing you can do to break the draft, other than be far enough ahead so that it's not a factor. You should know by the end of the race whether the draft effect is enough to allow you to pass the car ahead by the finish line.

"The final laps of a close race? You need just incredible focus and concentration, not trying to force the car into something it doesn't want to do. Not overdriving the car, but you have to be perfect, running it as hard as you can. For me that means getting super focused and shutting everything else out completely. You have to think about how you're going to drive the car and not how great you're going to look on the podium. The 'how' carries the win along with it. You can't waste your energy wishing for it to happen."

—Jeremy Dale

During the course of a close race you should try a number of draft-bys and see where it puts your car relative to the other as you cross start/finish. You'll also get an appreciation for whether your opponent has a draft or horsepower advantage on you. Swapping places early in the race can let you see if your rival's passes get completed before or after yours do.

Knowing the acceleration potential of your car helps form your last lap strategy. If your car is a little stronger, but on a normal run off the last corner the other driver is still able to squeak by at start/finish, you may be able to use your acceleration advantage to thwart his last lap attempt. A good draft-by pass starts with the second car in line having room ahead to accelerate into. With slightly more horsepower, you could lead into the last corner but take it a little slower than normal and close up the distance between you and the car behind. You have to do this in the corner, where the other driver is forced to stay on-line behind you. Once he's tight behind, the other driver can't accelerate away from the corner until you do; and if you have a slight acceleration advantage, your rival won't be able to get by at the flag.

If you're following, you may find yourself breathing out of the throttle on the approach to the last corner to keep sufficient spacing between you and the driver ahead. It can turn into a real game of cat and mouse with the lead driver slowing to impede the driver behind and the following driver slowing further to leave space ahead. At some point, the leading driver may slow so much that the driver behind will

switch tactics and accelerate hard, trying to pass on the way into the corner. Sometimes the third driver in line is the true beneficiary of all this slowing down.

If the draft isn't the issue, the relative placement of two closely-matched cars often comes down to who's ahead at the end of the last of the two or three best passing places on the racetrack—corners which require a significant speed loss. If the braking zone is relatively long, there is a chance that, even without a significant draft advantage, one driver can outperform the other in braking to an extent that he or she can pull alongside by the turn-in point. On the last lap, the lap time doesn't matter nearly so much as the relative positions of the cars, and a late-braking maneuver, although it might make both cars slower overall, can accomplish the goal of changing positions.

Defense

If out-braking is the issue, the lead car can defend by making it tougher for the opponent to pull it off. It's much harder to brake late and beat the car ahead to the apex if the lead car is on the inside of the road at the approach to the corner.

On the inside, it's a shorter route to the apex and easier to get there first, consequently taking the preferred line away from the car on the outside. It's possible to make the outside out-braking maneuver, but it's so uncommon that just your presence on the inside may dissuade the other driver from trying it.

There's nothing dirty about driving up to a corner on the inside of the road and denying your pursuer the preferred out-braking path, but you have to choose this path before the car behind makes a move toward it. If you drive in your mirrors, intentionally driving to get in the way of the path the driver behind has chosen, you're stealing your position in the race, not earning it, and you deserve the contempt you will no doubt get from your competitors.

"I'll never pull over on a guy if he's already committed. I'll run down the middle of the racetrack or on the inside approach to a corner, but as far as forcing the other driver to make a reactive move—I just don't do it. I grew up thinking 'I'm going to give you a certain amount of pavement, you've got to give me a certain amount of pavement.' You don't have to be a jerk. I pride myself on bringing the car in after a six- or twelve-hour race without a scratch on the car, and maybe coming in with a checkered flag.

"It's okay to drive a different line. If you can alter or confuse the other driver's game plan, that's all fair. But the bobbing and weaving stuff just doesn't belong."

—Terry Earwood

CARING FOR THE CAR

The driving, especially in tight traffic, will use up a great deal of your conscious attention, but you also have to make certain that your engine is not going to fail you. At least once a lap, you need to look at the water temperature and the oil pressure gauges to see if the motor is staying within its parameters. It takes less than a second to peek at the gauges—time you can afford to spend, especially considering the cost to repair a blown motor caused by running it too hot or low on oil pressure. The easiest place to do this is on a long straight after you have shifted to top gear. Work your way from the left mirror to the gauge furthest on the left and continue on from left to right surveying the gauges until you finish off the drill by looking in the right hand mirror. Make a routine of checking gauges by doing it on the upshift to top gear once a lap, and eventually it becomes automatic, leaving more time for conscious decisions.

Sterling Moss claimed to have developed the ability to read the gauges with his peripheral vision, never having to glance down from the road ahead to study their readings. So that there is no need to consciously try to read a numerical value off the gauge, it's helpful to paint a line on the surface of the gauge at its critical value: a minimum value for oil pressure, a maximum value for water temperature. If you see the needle on the gauge on the wrong side of the painted line, you can then decide to take a closer look to see specifically what value you're at.

Traffic and Cooling

If the race is a tight one and you end up running close behind another car or cars for many laps, the lack of clean air may affect your car's cooling ability. This goes for both engine coolant temperature and the temperature of the brakes. Without a straight shot of undisturbed air into the radiators, the engine temperature often rises. You may have to move out of line on

"At races where the ambient temperatures are high and you don't have long straights to get a lot of air to the brakes—even on something like an Indy Car, which generally has great brakes—they can overheat. What you do is brake a little longer and lighter, carrying the same entry speed to the corner, but taking a longer period of time to do it. You can't race someone like that because they'll eat you alive in the braking zone, but you can't just retire. Once you're aware of the problem, you reach over with your left foot and tap the brake pedal as you're driving down the straight to make sure it's up. Hello? Anybody there?"

—Brian Till

the straights in order to get a fresh charge of air into the radiator to keep your car from overheating. This influences your decision to lead or follow in your individual duel with another competitor. If your particular car has excess cooling capacity, traffic probably won't be an issue; but it always seems that racecars following too closely for too long will push the engine temperature over the edge.

The same goes for brakes. They need an ample flow of air over the rotors and calipers to keep the brake fluid under its boiling point. Reduce the air flow by following in the draft, and the fluid may begin to vaporize, making the pedal feel progressively spongy and limiting the squeezing capability of the caliper.

If the brakes are overheating, it's usually a progressive thing. Normally, the caliper with the worst air flow to it will go first. If it's a rear caliper, you'll feel more travel and sponginess in the brake pedal, followed by front-tire lock-up. Since you have a non-functioning rear brake, the fronts pick up the extra braking load lost by the rear, and this overload exhibits the same symptom as too much front brake bias. If a front caliper boils first, you'll feel sponginess in the pedal and a rear brake bias—much tougher to handle, since the rear end of the car will want to step out in either direction at the slightest provocation.

If the brake pad material goes beyond its optimum temperature before the brake fluid boils, you'll feel a loss in stopping power but the pedal won't feel spongy. If the brake pedal is rock hard but the car doesn't seem to slow like it used to, you're probably overheating the pad. They seldom go all at once, so you're likely to feel a change in brake bias too.

You can try to nurse the brakes back to health by getting into some clean air on the straights. You also have the option to take some strain off of the brakes

by braking earlier and more lightly and using downshifting to help the deceleration process. If the fluid has boiled, the pedal will never come back to being as hard as it was at the start. You can help make it better by pumping the brake pedal as you approach the braking zones. This compresses the vapor in the lines and calipers so that when you finally do go to the brakes in earnest, the fluid has a smaller pocket of vapor to compress before it applies pressure to the pucks and pads.

If the problem is overheated pad material, cooling the brakes down will restore the original braking ability of the car; but as they cool, the fronts and rears are likely to come back at different rates and the brake bias is likely to change as the process goes on.

POST-RACE CRITIQUE

The process of self-critique is one which goes on continuously as you drive the car. To develop as a driver, it is very useful to evaluate your performance immediately following a race.

We all fall into the trap of judging our progress purely by finishing position. The reality is that, in certain fields, you can finish high up but still have done a less-than-stellar job as a racer. To progress, you should take stock of the aspects of your racing technique which only you know about—be perfectly honest with yourself about your performance. Whether you use a written or mental checklist, evaluate your performance in each of the following areas and you can review your shortcomings before the next event in order to work on improving them.

- 1) Pre-pace prep: car and driver
- 2) Pace lap
- 3) Start
- 4) Line
- 5) Corner exit and car control
- 6) Braking
- 7) Shifting: up and down
- 8) Reading car changes
- 9) Routine for gauge-checking
- 10) Use of mirrors
- 11) The broad view
- 12) Concentration lapses
- 13) Approach to going faster
- 14) Passing technique

It's seldom that any of us, even after winning a race, can give ourselves high ratings in all categories. If you have a measure of success, however, you're likely to take on the next level of competition; and it gets tougher and tougher as you get closer to the top. This kind of post-race self-analysis helps keep you honest and gives you a chance to recognize mistakes so that you can meet the challenge of faster drivers.

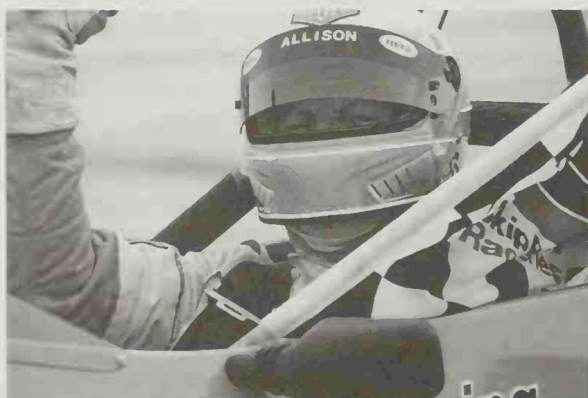


Fig. 10-8. Nothing beats winning, but avoid the trap of grading your performance simply by your finishing position, especially when you're just starting out.